



COURSE HANDOUT

Date: 03/08/2026

In addition to part I (General Handout for all courses appended to the Time table), this portion gives further specific details regarding the course.

Course Code : EEE G549
Name of the course : Fundamentals of Optoelectronics
Instructors : Prof. Rahul Kumar (IC), Prof. Navneet Gupta, Prof. V. K. Chaubey
Lab Instructor : Deepak Pareek

I. Course Description:

This course introduces the principles, design, and applications of optoelectronic devices. The course begins with a description of the interaction of light with semiconductor materials in a p-n junction configuration. This includes the phenomena of absorption, electroluminescence, and stimulated emission. The distinction between direct and indirect semiconductor materials is noted. Basic devices are then described: photodiodes, light-emitting diodes (LEDs), semiconductor optical amplifiers, and laser diodes. Array detectors, including complementary metal-oxide-semiconductor (CMOS) and charge-coupled devices (CCD) arrays, and array LEDs are then introduced. Basic specifications and applications of each of these devices are described, including solar cells, imaging with array detectors, and LED displays.

II. Textbooks and References:

TB1: Optoelectronics and Photonics: Principles and Practices (2nd Edition) by S.O. Kasap.

R1: Solid State Electronic Devices by Ben Streetman & Sanjay Banerjee.

R2: Semiconductor Optoelectronic Devices by Pallab Bhattacharya.

III. Course Objectives:

- Understand the fundamental solid-state physics governing light-matter interactions in semiconductor materials.
- Analyze the operation, design parameters, and performance metrics of modern optoelectronic emitters (LEDs, Lasers) and detectors (Photodiodes, Solar Cells).
- Evaluate the engineering principles behind advanced display architectures, flexible electronics, and matrix-based image sensors.
- Apply theoretical knowledge through hands-on computational modeling and simulation of optoelectronic device behavior.





IV. Course Contents

Sl. No.	Module	Topic	Outcome(s)
1	Physics of semiconductors (6 hours)	Review of semiconductor concepts, carrier transport, and energy bands. Direct vs. Indirect bandgap semiconductors E-k diagrams). P-n junction principles, heterojunctions, and energy band diagrams under bias.	Map energy band diagrams for direct and indirect bandgap semiconductors using E-k diagrams to predict radiative efficiency.
2.	Light-Emitting Diodes (LEDs) (8 hours)	Core principles of electroluminescence, efficiencies, and output spectrum. High-intensity quantum well LEDs and modern materials. White LEDs, phosphors, and driver electronics, Organic LEDs (OLEDs), Quantum Dot LEDs (QLEDs)	Explain the role of quantum wells and modern semiconductor materials in high-brightness LED configurations. Evaluate the operating mechanisms of phosphor-converted white LEDs, OLEDs, and QLEDs for display and lighting applications.
3	Laser Diodes & Optical Amplifiers (6 hours)	Laser oscillations: threshold gain coefficient and gain bandwidth. Steady-state semiconductor rate equations. Single-frequency lasers and Vertical Cavity Surface Emitting Lasers (VCSELs). Semiconductor Optical Amplifiers (SOAs).	Formulate and solve steady-state rate equations to determine photon and carrier densities. Critique the design and structural benefits of single-frequency lasers, VCSELs, and Semiconductor Optical Amplifiers (SOAs).
4	Flexible substrates and display devices (6 hours)	Flexible silicon and SoI wafers, Techniques for fabricating flexible Si-substrates, Thin Film Transistors (TFTs), TFT-LCD Displays	Identify engineering constraints and candidate materials for flexible, stretchable, and wearable substrates. Assess the mechanical and electrical reliability of optoelectronic materials under mechanical stress.
5	Photodetectors & Photovoltaics (6 hours)	Principles of the p-n junction, p-i-n, and Avalanche Photodiodes (APDs). Schottky junction and photoconductive detectors. Photovoltaic devices and solar cell efficiencies. Noise parameters in photodetection.	Assess noise contributions (shot noise, thermal noise) to determine the Signal-to-Noise Ratio (SNR) and Noise Equivalent Power (NEP) of a detector. Analyze the photovoltaic effect and extraction efficiency parameters that dictate solar cell performance.





6	Image Sensors (6 hours)	Charge-Coupled Devices (CCDs). Active Matrix Arrays and Complementary Metal-Oxide- Semiconductor (CMOS) image sensors.	Design-map the architectural differences, power consumption, and fill-factors of Active Matrix Arrays and CMOS image sensors. Contrast CCD and CMOS technologies regarding frame rate, noise, and integration scalability.
7	Practical / Mini Project (4 hours)	Simulation of Optoelectronic devices.	Utilize device simulation software to model electrical and optical characteristics of optoelectronic components.

V. Evaluation Scheme and Schedule:

Evaluation Components	Duration	Weightage (%)	Marks	Date & time	Remarks
Mid-semester Test	90 min	25	25	10/10 FN2	CB
Lab Component	-	30	20	-	OB
Quizzes	10-15 mins	10	10	Total 5	CB
Comprehensive Examination	3 Hours	35	35	15/12 AN	CB/OB
TOTAL			100		

V. Tentative List of Experiments

Exp. No.	Experiment details
E1	Introduction, safety protocol (goggle verification), and basic apparatus review.
E2-3	Light-Emitting Diodes I-V, I-P curves, and spectral output mapping
E4-5	Diode LASERS: Threshold characterization, power monitoring, and wavelength profiling.
E6	Photoconductors and spatial optoelectronic light detection
E7	p-i-n photodetector dynamics and Solar Cell power generation efficiency.

VII. Chamber Consultation Hour: Every Monday 5-6 PM.

VIII. Notices concerning the course: All notices concerning the course will be communicated via Nalanda.

IX. Make-up Policy: Make up will be permitted only in genuine cases with prior permission. Other than medical reason prior permission of IC is compulsory. No make up for quizzes.

Instructor-in-Charge

